ECI v7.5 amendment: NPP20 lepton sector and G1.12.B SU(5) modular $f^{ij}(\tau = i)$ quark sector, with Karam–Palatini Cassini reference, King–Wang dS-trap mechanism, and CSD$(1 + \sqrt{6})$ leptogenesis

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Draft: 2026-05-05 (night, post wave 4-5 and late-evening grafts)

Abstract

The v7.4 amendment of the Extended Cosmological Index (ECI) research programme [**?**] relaxed the strict S-fixed point axiom $H_5 \to H_5'$ (CM-anchored attractor) and replaced the Damerell–Coates–Sinnott bridge $H_7 \to H_7'$ by a two-part bridge (integer-point Damerell ladder + BDP p-adic complement). The present v7.5 amendment records six concrete extensions, one retraction, and one structural reframing.

Retraction (A46): the v7.4 "LYD20 unified scaffold with τ pinned at $\tau_S = i$" graft is empirically REFUTED. A 19-observable joint quark+lepton refit (PDG 2024) yields χ^2 penalty $\approx 99{,}061\times$ versus the LYD20 published best fit at $\tau_{\mathrm{LYD20}} = -0.21 + 1.52\,i$ – five orders of magnitude beyond the $\sim 1.8\times$ tolerance estimated by the earlier A42 scoping. v7.5 therefore adopts Option C: NPP20 lepton-only at $\tau_S = i$ (4–6 free parameters for 9 observables) plus the G1.12.B SU(5) $5_H + 45_H$ modular $f^{ij}(\tau = i)$ campaign for the quark sector, DROPPING the unified scaffold.

Grafts: (1) a stand-alone Cassini-wall section grafting the Karam–Sánchez López–Terente Díaz analytical PPN reference (KSTD 2026, `arXiv:2604.16226`), with v7.5-P Palatini sub-branch registered as a conditional 2027-decision direction; (2) the King–Wang multi-modulus dS-trap (KW 2023, `arXiv:2310.10369`) supplying a string-cosmological mechanism for $\tau \to i, \omega$ vacua (closes A14's "WHY $\tau = i$?" gap); (3) the dMVP26 canonical-Kähler quark hierarchy at $\tau = i$ (A48 PARTIAL graft) with $J^q_{CP} = 0$ cost flagged; (4) leptogenesis from the CSD$(1 + \sqrt{6})$ Littlest Modular Seesaw at $\tau_S = i$ (A55) yielding the structural signature $(n-1)^2 = 6$ in ϵ_1 scaling and Planck-compatible $Y_B^{\mathrm{calib}} = 0.872 \times 10^{-10}$.

Structural reframings: (5) the A42 "two-τ is $60\times$ worse" claim is corrected per A49 autopsy of Du–Wang `arXiv:2209.08796` – the comparison was a category error; two-τ is GENERICALLY HELPFUL empirically. v7.5 keeps single-τ on STRUCTURAL grounds (Du–Wang's GUT matter-unification orbifold-BC reduction), not numerical χ^2. (6) The A54 BC×CM at $\beta = 2\pi$ chain (Connes–Marcolli–Ramachandran 2005, `arXiv:math/0501424`) is honestly downgraded: it does NOT bypass H_6 (χ_4 nebentypus restriction) and joins the Galois-descent chain (P-NT) as a SECOND H_6-co-dependent partial derivation, not two independent analytical derivations.

We collect seven experimental disambiguators 2027–2045 (NA62, JUNO, DUNE, KamLAND-Zen, CMB-S4, Belle II, Hyper-K) in Section **??** and list open questions in Section **??**. The arithmetic-modular core ($f_{4.5.\mathrm{b.a}}$, $\mathbb{Q}(i)$, Damerell ladder, CSD$(1 + \sqrt{6})$, G1.12.B M1–M6) is preserved across the v7.4 $\to$ v7.5 transition.

1 Introduction: from v7.4 to v7.5

The v7.4 amendment [?] closed three open questions of the v7.3 axiomatic restart of ECI [?]: it relaxed the strict S-fixed point H_5 to a CM-anchored attractor H_5' ($|\tau - i| \lesssim 0.2$, supported by a 30×30 χ^2 scan W1 yielding $\tau^* = -0.1897 + 1.0034\,i$ at $\chi^2/\mathrm{dof} = 1.05$); replaced the half-integer Damerell–Coates–Sinnott bridge H_7 – refuted by the G1.15 audit on parity grounds for an odd-weight CM newform – by an integer-point Damerell ladder H_7' ($\alpha_m \in \{1/10, 1/12, 1/24, 1/60\}$ for $m \in \{1, 2, 3, 4\}$) with parameter-free Cardy hits $\rho_{\mathrm{Cardy}}(c = 1) = 1/12$ (free boson) and $\rho_{\mathrm{Cardy}}(c = 1/2) = 1/24$ (Ising); and recorded six wave 4–5 follow-ups (A14, A16, A17, A22, A24, A26).

The present v7.5 amendment records the post wave 4–5 development of 2026-05-05 (evening and night), comprising one retraction (A46), four grafts (A47, A48, A50, A55), and two structural reframings (A49, A54). Section **??** grafts the Karam–Sánchez López–Terente Díaz analytical PPN reference for the Cassini wall (KSTD 2026, `arXiv:2604.16226`) and registers a Palatini sub-branch v7.5-P as a conditional 2027-decision direction. Section **??** collects the v7.5 modular flavour body in Option-C form (NPP20 lepton + G1.12.B quark), with subsections covering: G1.12.B SU(5) M1–M6 closure including the M5 partial-width computation (Sec. **??**); the CSD($1 + \sqrt{6}$) Littlest Modular Seesaw at $\tau_S = i$ (Sec. **??**); the A24 $\sqrt{6}$ Galois-rationality closure (Sec. **??**); the A47 King–Wang dS-trap physical mechanism for $\tau = i, \omega$ (Sec. **??**); and the A55 leptogenesis $(n-1)^2 = 6$ structural signature (Sec. **??**). Section **??** records the A48 partial graft of the dMVP26 canonical-Kähler quark hierarchy at $\tau = i$ with the $J^q_{CP} = 0$ cost. Section **??** re-frames the two-τ debate per A49. Section **??** preserves the v7.4 sub-algebra H_1 closure, Damerell ladder, and A17 numerological CKM hits. Section **??** records the conditional analytical τ derivations (A54 BC×CM and Galois descent in P-NT) honestly framed as H_6-co-dependent. Section **??** lists seven experimental disambiguators 2027–2045. Section **??** collects open questions and the v7.5-P Palatini sub-branch decision matrix.

A46 retraction (the leading v7.4 → v7.5 change). The v7.4 strategic synthesis (commit `6.0.53.5`) had assumed A42's scoping estimate that the LYD20 unified quark+lepton scaffold of [?], when τ is strictly pinned at the S-fixed-point $\tau_S = i$, would carry an empirical χ^2 penalty of order 1.8× versus LYD20's published best fit at $\tau_{\mathrm{LYD20}} = -0.2123 + 1.5201\,i$. A46 (this volume) ran the explicit 19-observable joint Q+L refit (10 free LYD20 parameters after pinning τ):

Fit configuration	τ	$\chi^2_{\min}$ (19 obs)	Penalty
LYD20 published best	$-0.2123 + 1.5201\,i$	**1,009**	reference (0×)
W1 attractor (v7.4)	$-0.1897 + 1.0034\,i$	878,990	+870×
Strict $\tau_S = i$ (v7.4 pinning hypothesis)	$0 + 1.0\,i$	**$\sim 10^8$**	**+99,061×**

Table 1: A46 single-τ refit of the LYD20 unified Q+L scaffold, PDG 2024 covariances, 19 observables (8 fermion masses + 4 CKM moduli + 4 PMNS angles + δ_{CP} + 2 mass-squared splittings). The v7.4 "pin τ at i" hypothesis is REFUTED at ~ 5 orders of magnitude beyond the A42-scoped tolerance. Full results JSON in `notes/eci_v7_aspiration/A46_LYD20_GRAFT/lyd20_pinned_results.json`.

The catastrophic penalty traces to the V2 no-go mechanism [?]: at $\tau_S = i$ the $\{d^c, s^c\}$-block of the down-quark Yukawa becomes near-rank-1 by an S-collinearity lemma, killing $\sin\theta_C$ and propagating into PMNS through the LYD20 unified-scaffold's structural quark–lepton mixing.

Honest revision (Option C of A46). v7.5 abandons the unified-scaffold pinning and adopts:

- *Lepton sector:* NPP20 model [?] ($S_4' \hookrightarrow \widetilde{\Gamma}(4)$ doublet/triplet assignments, 4–6 free parameters for 9 observables) at the strict S-fixed point $\tau_S = i$, in CSD($1+\sqrt{6}$) Littlest Modular

Seesaw alignment.

- *Quark sector:* G1.12.B SU(5) $5_H + 45_H$ modular construction with $f^{ij}(\tau = i)$ built from row-by-row LYD20-Model-VI contractions (rows u, c, t at weights $1, 2, 5$, irreps $\hat{3}'$, $\mathbf{3}$, $\hat{3}$); modular invariance inherited; M1–M2 closed (this volume) and M3–M6 forecast at 3.5–5 sub-agent months. Quark hierarchies via dMVP26 canonical-Kähler at $\tau = i$ (Section **??**).

The arithmetic anchor $f_{4.5.b.a}$ (LMFDB CM newform 4.5.b.a, weight 5, CM by $\mathbb{Q}(i)$) and the integer-point Damerell ladder H'_7 are unchanged across the A46 retraction. The W1 attractor result (v7.4) is now interpreted as a partial-fit structure on the 7 fermion mass ratios alone, NOT a unified Q+L claim.

A49 correction propagated. The A42 row-5 claim "two-τ is 60× worse" was a category error: it compared the single-τ Sample IX′ sweet spot ($\chi^2_{\rm tot} = 1.58$) of Du–Wang Scenario III against the dual-τ generic-bad worst sub-scenarios ($\chi^2 \approx 95$). Du–Wang [**?**, Tab. 5, §4] themselves report two-τ *improves* the lepton fit 4–16× in every sub-scenario where they tabulate both. Section **??** reframes the v7.5 single-τ choice as a STRUCTURAL preference (GUT matter unification, Du–Wang's $M_4 \times S^1/Z_2$ orbifold-BC reduction) rather than an empirical χ^2 argument.

Hallucination policy. This amendment reuses the v74_amendment_v2 verified bibliography verbatim (28 entries, all live-verified 2026-05-05 via the arXiv API) and adds 8 new live-verified entries for the v7.5 grafts (KSTD26, KW23, KMSR18, CMR05, Bertotti03, KZ24, CMBS4, JUNO22). The hallucination counter is held at 85. Author-name errata respected throughout. Mistral large-latest STRICT-BANNED. Computational substrate: PC RTX 5060 Ti (cosmopower-jax 386 predictions/s, JAX named_shape patch); LMFDB queries cached 2026-05-05.

2 Cassini wall and Palatini formulation

The NMC branch of ECI v7.4 enforces a Cassini compliance wall $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 6 \times 10^{-6}$, recovered from the standard Solar-System bound $|\gamma - 1| \leq 2.3 \times 10^{-5}$ (Cassini Doppler tracking, [**?**]) translated into the Jordan-frame Lagrangian $\mathcal{L} \supset -\frac{1}{2}\xi_\chi R\chi^2$ (Faraoni convention; cf. [**?**], App. A). In this section we graft an explicit analytical PPN reference for the wall and register a Palatini sub-branch (ECI v7.5-P) as a conditional 2027-decision research direction.

Analytical Cassini-wall reference (KSTD 2026). Karam, Sánchez López, and Terente Díaz [**?**] (`arXiv:2604.16226`) recently derived a unified post-Newtonian treatment of scalar–tensor gravity with arbitrary non-minimal coupling $A(\Phi)$, kinetic function $B(\Phi)$, and potential $V(\Phi)$, in BOTH the metric and Palatini formalisms. In their notation, with $\delta_{\rm P} = 0$ (metric) or $\delta_{\rm P} = 1$ (Palatini) and A_n, B_n, V_n the Taylor coefficients of A, B, V about the cosmological background, the effective scalar mass and PPN parameter γ at distance r from a point source read

$$m_\varphi^2 = \frac{16\pi G\, A_0\, V_2}{16\pi G\, A_0\, B_0 + 3A_1^2(1-\delta_{\rm P})}, \tag{KSTD 3.38}$$

$$G_{\rm eff}(r) = \frac{G}{A_0}\left[1 + \frac{A_1^2\, e^{-m_\varphi r}}{16\pi G\, A_0\, B_0 + 3A_1^2(1-\delta_{\rm P})}\right], \tag{KSTD 3.41}$$

$$\gamma(r) = 1 - \frac{2A_1^2\, e^{-m_\varphi r}}{16\pi G\, A_0\, B_0 + 3A_1^2(1-\delta_{\rm P}) + A_1^2\, e^{-m_\varphi r}}. \tag{KSTD 3.42}$$

Specialising to a non-minimally coupled scalar with canonical kinetic term ($B_0 = 1$, $B_1 = 0$) yields the metric Yukawa coupling $\alpha = A_1^2/(16\pi G A_0 + 3A_1^2)$ and the Palatini one $\hat{\alpha} = A_1^2/(16\pi G A_0)$ (KSTD eqs. 4.31–4.32); the analogous expression for $\beta(r)$ is given by KSTD eq. (3.67).

Consistency check with the ECI v7.4 Cassini wall. For the ECI Lagrangian $-\frac{1}{2}\xi_\chi R\chi^2$ expanded about χ_0, one identifies $A_0 = 1-\xi_\chi\chi_0^2/M_{\rm P}^2$ and $A_1 = -\xi_\chi\chi_0/M_{\rm P}$, so that $A_1^2/(16\pi G A_0) \simeq 4\,\xi_\chi^2(\chi_0/M_{\rm P})^2$ to leading order. In the metric formalism with strong Yukawa suppression at Solar-System scale, KSTD eq. (3.42) reduces to $|\gamma - 1| \approx 2A_1^2/(16\pi G A_0) \simeq 8\xi_\chi^2(\chi_0/M_{\rm P})^2$, and the Cassini bound $|\gamma - 1| \leq 2.3 \times 10^{-5}$ translates into $\xi_\chi^2(\chi_0/M_{\rm P})^2 \lesssim 3 \times 10^{-6}$, i.e. $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 6 \times 10^{-6}$ at the small-ξ_χ root — in agreement with the wall enforced by the v7.4 NMC posterior [**?**, App. A]. The ECI v7.4 mean $\xi_\chi = -3 \times 10^{-5}$ (Levier 1B) sits $\sim 10^2$ orders of magnitude inside the wall, so the NULL at Cassini is structurally enforced rather than fine-tuned.

Forward-pointing remark: the v7.5-P Palatini sub-branch. KSTD show explicitly that, at fixed (A_0, A_1, B_0), the Palatini formalism admits a substantially wider allowed region in V_2 than the metric one — by up to ~ 10 orders of magnitude in the regime of small A_0 and large $|A_1|$ (KSTD Fig. 2 and §4.2.1). The mechanism is geometric: in Palatini the screening length is $\hat{m}_\varphi^{-1} = V_2^{-1/2}$, independent of $A_0\,B_0$, leading to stronger Yukawa suppression of the fifth-force term in $\gamma(r)$. As an open research direction, an *ECI v7.5-P Palatini sub-branch* (sketch in Section **??**) may relax the Cassini wall by several orders of magnitude while preserving the cosmological NMC signal at $\xi_\chi \sim O(10^{-2})$ – re-opening contact with detection-side analyses such as [**?**] that lie outside the current ECI metric-formalism wedge. The arithmetic-modular anchors of v7.4–7.5 ($f_{4.5.b.a}$, τ-attractor, Damerell ladder, CSD$(1+\sqrt{6})$, G1.12.B) are formalism-independent and carry over unchanged. *We do not commit to v7.5-P in the present amendment*; the metric formalism remains the conservative baseline, and the Cassini NULL falsifier listed in Section **??** applies to v7.5-metric only.

3 Modular flavour body (Option C: NPP20 lepton + G1.12.B quark at $\tau = i$)

Per the A46 retraction (Section **??**), the v7.5 modular flavour body splits into: a lepton sector following NPP20 [**?**] at the strict S-fixed point $\tau_S = i$, in CSD$(1+\sqrt{6})$ Littlest Modular Seesaw alignment (Section **??**); and a quark sector following the G1.12.B SU(5) $5_H + 45_H$ modular construction (Section **??**) with hierarchies supplied by dMVP26 canonical-Kähler at $\tau = i$ (Section **??**). The A47 King–Wang multi-modulus dS-trap supplies the string-cosmological mechanism that selects $\tau \in \{i, \omega\}$ at the level of the modulus potential (Section **??**); the A14 cross-K test then selects $K = \mathbb{Q}(i)$ uniquely.

3.1 NPP20 lepton sector at $\tau_S = i$

The lepton sector follows Novichkov–Penedo–Petcov [**?**]: S_4' doublet/triplet assignments for the three left-handed lepton generations, two heavy right-handed singlets, and the type-I seesaw with $\tau_S = i$ pinned by the residual $\mathbb{Z}_4 \subset \mathrm{SL}(2,\mathbb{Z})$ symmetry at the S-fixed point. The model has 4–6 free parameters for the 9 PMNS observables ($\theta_{12,13,23}$, δ_{CP}, $\Delta m^2_{12,23}$, Σm_ν, $m_{\beta\beta}$, and one high-energy Majorana phase η). At $\tau_S = i$ the weight-2 S_4' triplet form $Y_3^{(2)}$ acquires the alignment $(1,\ 1+\sqrt{6},\ 1-\sqrt{6})$ (DKLL19 Table 1, Case B, [**?**]), which feeds the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw studied in Section **??**.

3.2 G1.12.B SU(5) modular $f^{ij}(\tau = i)$ campaign: M1–M2 PASS, M5 partial-width computation

The G1.12.B campaign [**?**] is the v7.4–7.5 proton-decay falsifier programme: a six-milestone modular SU(5) $5_H + 45_H$ construction targeting a parameter-free branching ratio

$$\frac{B(p\to e^+\pi^0)}{B(p\to K^+\bar{\nu})} \in [0.3,\, 3] \quad (95\%\ \text{CL post-M}_6),$$

to be tested by Hyper-K + DUNE 2030+ data. The published v7.4 amendment recorded M1+M2 PASS; the present v7.5 amendment adds the M5 partial-width computation result. M3–M6 forecast is 3.5–5 sub-agent months at \$0–100.

M1 (A22) PASS 4/4. The 45-of-SU(5) Higgs is assigned to the S_4' trivial singlet $\mathbf{1}$ at modular weight $k_{45} = 0$, with the off-diagonal Yukawa form-factor $f^{ij}(\tau)$ built from row-by-row LYD20-Model-VI contractions (rows u, c, t at weights $1, 2, 5$, irreps $\hat{3}', \mathbf{3}, \hat{3}$). The 3×3 matrix $f^{ij}(\tau = i)$ is computed symbolically (sympy) and numerically; all 6 off-diag entries are non-zero, modular invariance under S and T is verified to machine precision (residual $\leq 10^{-15}$), and the A2 +19.5% closure check matches at +19.43% (within 0.07 pp). The non-modular precedent is Haba–Nagano–Shimizu–Yamada [**?**].

M2 (A26) PASS 5/5. The off-diagonal up-quark Yukawa SVD at the W1 attractor $\tau^* = -0.1897 + 1.0034\,i$ reproduces the Antusch–Hinze–Saad GUT-scale targets [**?**] ($y_t = 0.4454$, $y_c/y_t = 2.725 \times 10^{-3}$, $y_u/y_c = 2.05 \times 10^{-3}$) to machine precision: the fit $(\beta_u/\alpha_u, \gamma_u/\alpha_u) = (152.42, 305.38)$ achieves $\chi^2_{\text{up}} = 6.7 \times 10^{-20}$ (2 parameters, 2 observables, exact); $|\Delta(y_c/y_t)| = 10^{-9}\%$. Unitarity of U_L, U_R holds at $\sim 5 \times 10^{-16}$; SVD reconstruction at $\sim 7 \times 10^{-11}$.

M5 partial widths (A36, this volume). The full six-channel partial-width computation at the central $(M_{T_5}, M_{T_{45}}) = (10^{15}, 10^{12})$ GeV reference uses Haba eqs. (18)–(23) [**?**] (verified by direct PDF read), Aoki–Soni $\alpha_H(2\,\text{GeV}) = -0.0144\,\text{GeV}^3$ chiral parametrisation, FLAG-2024 lattice form factors [**?**], and the standard $D = 0.8$, $F = 0.46$, $f_\pi = 0.093\,\text{GeV}$, $A_{RL} = 2.6$, $\lambda_W = 0.22$ parameters. Six channels $\{p \to e^+\pi^0, \mu^+\pi^0, \bar{\nu}K^+, \mu^+K^0, e^+K^0, \bar{\nu}\pi^+\}$ are evaluated against current Super-Kamiokande limits [**?**, **?**] and the Hyper-K 20-yr [**?**] / DUNE 20-yr [**?**] reach. Numerical verification of all input constants is recorded in `notes/eci_v7_aspiration/A36_G112B_M5/tau_p_results.json`. The decisive falsifier remains unchanged from v7.4:

> *ECI v7.5 modular SU(5) $5_H + 45_H$ at $\tau \sim \tau^*$ (and quark-sector hierarchy at $\tau = i$ via dMVP26 Kähler) predicts $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ (95% CL post-M$_6$). Hyper-K + DUNE combined 2030+ data outside $[0.3, 3]$ at $> 3\sigma$, OR $\tau(p \to e^+\pi^0) > 3 \times 10^{35}$ yr at $> 2\sigma \Rightarrow$ **REFUTED**.*

The post-M$_6$ central forecast is $\tau(p \to e^+\pi^0) \sim 3 \times 10^{34}$ yr, $\tau(p \to K^+\bar{\nu}) \sim 10^{34}$ yr – both above current Super-K limits and well within Hyper-K + DUNE 20-yr reach.

3.3 CSD$(1 + \sqrt{6})$ Littlest Modular Seesaw (A14)

A second probe of the $K = \mathbb{Q}(i)$ anchoring exploits the strict S-fixed point $\tau_S = i$ in the lepton sector. At $\tau_S = i$ the weight-2 S_4' triplet form $Y_3^{(2)}$ admits the alignment

$$Y_3^{(2)}(i) \;\propto\; \big(1,\; 1 + \sqrt{6},\; 1 - \sqrt{6}\big), \tag{1}$$

the so-called CSD$(1 + \sqrt{6})$ alignment of the Littlest Modular Seesaw [**?**, **?**]. Equation (**??**) drives a 2-real-parameter neutrino sector with two right-handed Majorana singlets, predicting normal ordering with $m_1 = 0$, Dirac CP phase $\delta_{CP} \approx -87°$, first-octant $\sin^2\theta_{23} \approx 0.46$–$0.55$, and

$\Sigma m_\nu \approx 0.06\,\mathrm{eV}$. The cross-$K$ test of DKLL19 Table 1 (A14) confirms $K = \mathbb{Q}(i)$ uniqueness: only at $\tau_S = i$ ($K = \mathbb{Q}(i)$) does the weight-2 S_4' triplet acquire the non-trivial nested-radical alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$; at $\tau_{ST} = \omega$ ($K = \mathbb{Q}(\sqrt{-3})$) the alignment collapses to a permutation vector $(0, 1, 0)$, killing the seesaw.

Falsifiers (Sec. ?? items 4–5). DUNE 2030+ at $\pm 15°$ on δ_{CP} rules out the prediction at $> 3\sigma$ if δ_{CP} centres at 0 or $+90°$. JUNO 2030 NMO at 3–4σ [**?**] independently kills the NO + $m_1 = 0$ prediction if inverted. KamLAND-Zen 2027 / nEXO 2030+ tests $m_{\beta\beta} \sim$ few meV.

3.4 $\sqrt{6}$ structure: A24 closure note

A PSLQ search at dps = 60, $|c| \leq 500$, in seven distinct bases $\{\sqrt{6}, \Gamma(1/4), \pi, \log 2, \log 3, \Omega_K, \log\theta_{2,3,4}(i), \log\eta(i)\}$ (A24) yields no Q-linear relation of $\log\sqrt{6}$ with the Chowla–Selberg periods of $\mathbb{Q}(i)$: in every non-trivial relation found, the coefficient on $\log\sqrt{6}$ is zero. The $\sqrt{6}$ in Eq. (**??**) is therefore a Galois-rational quantity (it is the discriminant $\sqrt{b^2 - 4c}$ of the secular 2×2 polynomial of $Y_3^{(2)}(i)$ eigenvalues), *not* a period-theoretic invariant of $\mathbb{Q}(i)$. Consequently the $K = \mathbb{Q}(i)$ anchor of CSD$(1+\sqrt{6})$ is *combinatorial* (S-fixed-point of S_4'), not a deep CM-period identity via Chowla–Selberg. This keeps the empirical falsifier (DUNE 2030+) valid but classifies the $1+\sqrt{6}$ alignment as a "thin coincidence" in the Lakatos sense, not a hard prediction lifted by deep number theory. Future probes: $\mathbb{Q}(\sqrt{-24})/\mathbb{Q}(\sqrt{-6})$ period rings (which contain $\sqrt{6}$ natively).

3.5 King–Wang multi-modulus dS-trap mechanism for $\tau = i, \omega$ (A47)

A14's empirical-deepness assessment ($P \sim 20\%$, "WEAK CONSTRAINT, deep") had flagged the absence of a physical motivation for $\tau_l = i$ from the modular-flavour literature. A47 closes this gap by grafting the King–Wang multi-modulus dS-trap of [**?**].

In the King–Wang setup, $\mathcal{N} = 1$ supergravity in the flavour modulus τ and dilaton S has superpotential

$$W \;=\; \Lambda_W^3\, \Omega(S)\, \frac{H(\tau)}{\eta^6(\tau)}, \qquad H(\tau) \;=\; (j - 1728)^{m/2}\, j^{n/3}\, P(j),$$

with $K \propto -\log(S + \bar{S})$ at tree level and Shenker-like $O(e^{-1/g_s})$ corrections. The key kinematic fact is that $\partial_\tau V$ has modular weight 2, so under the S-action at $\tau = i$ one has $(\partial_\tau V)|_i = (-i)^2 (\partial_\tau V)|_i$, forcing $\partial_\tau V \equiv 0$ at $\tau = i$. The same logic via ST (order 3) gives $\tau = \omega$. The Hessian is positive (KW Appendix C), so de Sitter minima emerge precisely at $\tau \in \{i, \omega\}$.

Compatibility with ECI v7.5. Three checks pass: (a) KW is purely modular-kinematic and makes no reference to CM points, Heegner values, or periods, so adding the ECI CM-by-$\mathbb{Q}(i)$ anchor is non-redundant. KW provides BOTH i and ω vacua; ECI's A14 cross-K test selects $\tau = i$ ($K = \mathbb{Q}(i)$) uniquely. (b) No double-counting: KW shapes $V(\tau, S)$, while the CM anchor selects *which* KW minimum is realised. (c) Cassini-clean: the KW τ is a SUGRA modulus with mass $m_\tau \sim m_{3/2}$ (TeV–10^{12} GeV), whereas fifth-force/Cassini constrains scalars with $m < 10^{-12}$ eV; this is a different scalar from the ECI NMC cosmological one and trivially evades the Cassini bound.

Empirical-deepness upgrade. A14's $P \sim 20\%$ ("WEAK, deep") is upgraded to $P \sim 35$–40% ("MEDIUM, deep") after the A47 graft, since the "WHY $\tau = i$?" question now has a string-cosmology answer (KW dS-trap) layered above the cross-K selection. Honest deficits remain: KW supply no numerical m_τ or Λ_W, no CM/class-field-theory connection (ECI fills unilaterally), and a Shenker-type uplift rather than KKLT-tuned dS (no strict no-tuning theorem).

3.6 Leptogenesis from CSD$(1+\sqrt{6})$ at $\tau_S = i$: the $(n-1)^2 = 6$ structural signature (A55)

The A14 graft (Section **??**) places the lepton sector at the modular fixed point $\tau_S = i$ of S'_4, where the weight-2 triplet alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$ yields the CSD$(1+\sqrt{6})$ Littlest Seesaw of King [**?**, **?**] with two right-handed neutrinos and $m_1 = 0$. This subsection records the implied baryon asymmetry of the universe (BAU) via flavoured thermal leptogenesis.

For CSD(n) Case A ($M_{\rm atm} = M_1 \ll M_{\rm sol} = M_2$), the hierarchical-limit decay-asymmetry parameters [**?**] read

$$\epsilon^A_{1,\mu} \simeq -\frac{3}{16\pi}\frac{M_1}{M_2}\, n(n-1)\, b^2 \sin\eta, \tag{2}$$

$$\epsilon^A_{1,\tau} \simeq -\frac{3}{16\pi}\frac{M_1}{M_2}\, (n-1)(n-2)\, b^2 \sin\eta, \tag{3}$$

whose flavour-summed total scales analytically as

$$\epsilon^A_{1,\rm tot} \;=\; -\frac{3}{16\pi}\frac{M_1}{M_2}\, 2(n-1)^2\, b^2 \sin\eta. \tag{4}$$

Substituting the ECI value $n = 1+\sqrt{6}$ from the modular fixed-point gives the exact rational signature

$$(n-1)^2 \;=\; (\sqrt{6})^2 \;=\; 6, \qquad \Longrightarrow \qquad \boxed{\frac{Y_B^{\rm CSD(1+\sqrt{6})}}{Y_B^{\rm CSD(3)}} \;=\; \frac{(n-1)^2}{(3-1)^2} \;=\; \frac{6}{4} \;=\; \frac{3}{2}}, \tag{5}$$

at fixed Yukawa amplitude b, high-energy Majorana phase η, and heavy Majorana masses $M_{1,2}$. The irrationality of $n = 1+\sqrt{6}$ *cancels* in $(n-1)^2$, leaving an integer factor 6 that mirrors the 3 in CSD(3) – a structural fingerprint of the modular $\tau_S = i$ alignment.

Calibrating to the King–Molina-Sedgwick–Rowley benchmark Case A2 ($M_1 = 5.05\times10^{10}$ GeV, $M_2 = 5.07\times10^{13}$ GeV, $a = 0.00806$, $b = 0.0830$, $\eta = 2\pi/3$), which gives $Y_B^{\rm CSD(3)} = 0.860\times10^{-10}$ at $\chi^2/{\rm dof} = 1.51/3$, Eq. (**??**) predicts

$$Y_B^{\rm CSD(1+\sqrt{6})}\Big|_{a,b,\eta,M_R\ \rm fixed} \;=\; 1.290\times10^{-10}, \tag{6}$$

which exceeds the Planck value $Y_B^{\rm obs} = (0.872\pm0.006)\times10^{-10}$ [**?**] by 48%. This is reabsorbed exactly by the rescaling $b^2\sin\eta \to (2/3)\, b^2\sin\eta$ (e.g. $b \to 0.0678$ at $\eta = 2\pi/3$), which leaves the low-energy PMNS parameters $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, \Delta m^2_{12}, \Delta m^2_{13})$ shifted by $\lesssim 1\sigma$ relative to the King Case A2 benchmark, well within the NuFIT 5.3 envelope. A viability scan returned 7/512 grid points inside the Planck $\pm1\sigma$ window, confirming non-trivial measure on the (b, η, M_R) space.

Falsifier status. ECI v7.5 does not promise a *parameter-free* BAU prediction in the A14/CSD$(1+\sqrt{6})$ sector – the high-energy phase η is independent of the low-energy Dirac phase $\delta_{CP} \approx -87^\circ$ in any two-RHN seesaw with one phase [**?**]. What v7.5 *does* predict is the structural signature $(n-1)^2 = 6$ in ϵ_1 scaling, testable indirectly via the joint $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, m_2, m_3, Y_B)$ $\chi^2/$dof fit on DUNE+JUNO+NuFIT 2030+ data. Honest probability assessment: graft into v7.5 successfully $\sim 55\%$, sharp falsifier added $\sim 10\%$, Foundational-Prize-class $\sim 3\%$.

4 Quark hierarchy via dMVP26 canonical Kähler at $\tau = i$ (A48 partial graft)

The G1.12.B SU(5) framework (Section **??**) supplies modular structure but consumes hierarchies. We graft de Medeiros Varzielas–Paiva [**?**] (`arXiv:2604.01422`) to supply the quark-mass hierarchy at $\tau = i$.

With S_4' weights $(k_Q, k_{u^c}, k_{c^c}, k_{t^c}, k_{d^c}, k_{b^c}) = (0, -6, -1, -10, -6, \ldots)$ (compatible with row-by-row LYD20 contractions of Section **??**) and Kähler metric

$$K \supset -\Lambda_K^2 \log[-i(\tau - \bar{\tau})] + \sum_i \frac{|\phi_i|^2}{[-i(\tau - \bar{\tau})]^{k_i}},$$

canonical normalisation absorbs $(2\,\mathrm{Im}\,\tau)^{-k_i/2}$ into the up-quark mass matrix M_u and down-quark mass matrix M_d, giving $y_u/y_c,\ y_c/y_t \sim 10^{-3}$ with real $\mathcal{O}(1)$ Yukawas $C_i \in [0.4, 5.6]$.

Survival at $\tau = i$ (A48 numerical check). For ECI v7.5 we PIN $\tau = i$ (CM-anchor at $K = \mathbb{Q}(i)$). The authors of [**?**] prove on p. 9 via analytical expansion $\tau = i + \epsilon$ that $J_{CP} \propto \mathrm{Re}(\epsilon)$ "completely decoupling the CP phase from the mass hierarchies, which are primarily related to $\mathrm{Im}\,\tau \sim 1$". The A48 SVD at $\tau = i$ versus the dMVP26 best fit $\tau^{\mathrm{dMVP26}} = 0.00455 + 1.00705\,i$ confirms shifts $< 5\%$ in the up-sector and $\sim 20\%$ in y_d/y_s:

Ratio	PDG-2024 (GUT)	dMVP26 best-fit	$\tau = i$ pinned (A48)
y_c/y_t	2.81×10^{-3}	2.99×10^{-3}	3.09×10^{-3}
y_d/y_s	5.00×10^{-2}	$2.27 \times 10^{-1}{}^*$	$1.81 \times 10^{-1}{}^*$
y_s/y_b	1.78×10^{-2}	1.78×10^{-2}	1.86×10^{-2}

Table 2: A48 SVD test of dMVP26 hierarchies under $\tau = i$ pinning (asterisk: a residual normalisation gap of order $\times 4$ in y_d/y_s relative to PDG affects both the best fit and the $\tau = i$ point and is traced to a likely Eq. (24) Clebsch–Gordan sign or weight-1 $\hat{Y}_3$ normalisation; the $\tau = i$/best-fit *ratio* is the relevant graft viability indicator and is $< 5\%$ in the up sector and $\sim 20\%$ in y_d/y_s). Authors' own analytic argument $J_{CP} \propto \mathrm{Re}(\epsilon)$ is the independent confirmation.

Cost: $J_{CP}^q = 0$ at strict $\tau = i$. At $\mathrm{Re}(\tau) = 0$ the CKM Jarlskog vanishes, so the CP-phase of CKM cannot come from the modular sector at strict $\tau = i$. v7.5 lists two compensator routes (open):

- *(a) Lepton-sector spurion:* the $\delta_{CP} \approx -87°$ of CSD$(1 + \sqrt{6})$ (Section **??**) feeds the CKM phase via a GUT-scale Yukawa-mixing term.
- *(b) RGE threshold* $\mathrm{Re}\,\tau \sim 10^{-3}$*:* a small renormalisation-group displacement of τ from i at the GUT-matching scale, of order $\mathrm{Re}\,\tau \sim 10^{-3}$, would give the observed Jarlskog $J_{CP}^q \sim 3 \times 10^{-5}$ via the linear scaling of [**?**] p. 9. This compensator is not yet derived from G1.12.B M3–M4 RGE matching; it is registered as a quantitative open question in Section **??**.

Both routes are compatible with the v7.5 Option-C architecture; neither is yet a parameter-free derivation. The A48 graft is therefore PARTIAL: hierarchies survive, CP phase is cost.

5 Two-tau MUDDY note: empirical helpfulness vs. GUT-unification preference (A42 + A49 correction)

The v7.4 strategic synthesis (commit `6.0.53.5`) included a row in A42's TOP-5 BSM comparison claiming that "two-τ models are 60× worse than single-τ". The A49 autopsy of Du–Wang [**?**] (`arXiv:2209.08796`, PDF directly read) corrects this as a category error: the "60×" arose from comparing the single-τ Sample IX$'$ sweet spot (Scenario III, $\chi^2_{\mathrm{tot}} = 1.58$) against Scenario III's generic-bad sub-scenarios in dual-τ ($\chi^2 \approx 95$, different sub-models). The correct table-by-table comparison is:

Sub-scenario (lepton sector only)	Single τ	Two τ
Scenario I (Du–Wang)	$\chi_l^2 = 236.3$	$\chi_l^2 = 48.8$
Sub-scenario IX (Scenario II)	$\chi_l^2 = 486.0$	$\chi_l^2 = 30.2$
Sub-scenario X′ (Scenario III)	$\chi_l^2 = 233.0$	$\chi_l^2 = 58.9$
Sample IX′ (Scenario III sweet spot)	$\boldsymbol{\chi^2_{\text{tot}} = 1.58}$	not reported

Table 3: A49 honest re-tabulation of Du–Wang Tab. 5: two-τ *improves* the lepton fit by 4–16× in every sub-scenario where both numbers appear. Sample IX′ wins because flipped SU(5) GUT-locks $Y^D = (Y^E)^T$ (one Yukawa makes both down-quark and charged-lepton matrices), so the lepton sector inherits any tuning that succeeds for quarks, plus the $(1', 1'', 1'')$ singlet assignments with weights $(2, 4, 2)$ for $\bar{F}_{1,2,3}$ provide enough freedom for one τ in the fundamental domain to satisfy quark hierarchy AND PMNS angles simultaneously. Mechanism is GUT-locked structure + lucky weight assignment, NOT cancellation.

Du–Wang's own theoretical objection to two-τ. Du–Wang [**?**, §4, conclusions] themselves argue that two-τ is theoretically problematic: GUT representations (flipped SU(5) $\bar{10}$, SO(10) **16**) contain BOTH quarks and leptons, so independent τ_q, τ_l "spoils unification of matter contents". They propose a 5D $M_4 \times S^1/Z_2$ orbifold that breaks $A_4^Q \times A_4^L \to$ diagonal A_4^D via boundary conditions (their eqs. 4.13–4.16), restoring a single low-energy modular symmetry.

v7.5 reformulation of the single-τ axiom H_5'. The A49 correction obliges us to drop the *empirical* "two-τ is 60× worse" argument for single-τ and replace it by a STRUCTURAL one:

> *The v7.5 single-τ choice is justified on UV matter-unification grounds (Du–Wang orbifold-BC reduction of $A_4^Q \times A_4^L \to$ diagonal A_4^D, generalised to S_4'). The CM-by-$\mathbb{Q}(i)$ anchor selects $\tau_S = i$ as the high-symmetry IR fixed point at which one modulus suffices. If v7.5 wants UV two-τ, the orbifold-BC reduction becomes a predicted constraint, not freedom.*

The A16 caveat of v7.4 (single-τ refit pulls $\sin^2\theta_{13}$ at 24.5σ under the LYD20 unified-scaffold embedding) is now acknowledged as confirmation of the A46 retraction: it was the LYD20 unified-scaffold embedding that broke under single-τ pinning, not single-τ itself. The v7.5 Option-C architecture (NPP20 lepton + G1.12.B quark) does not exhibit the A16 pull because NPP20 has independent lepton-sector freedom.

6 Sub-algebra H_1 closure, integer-point Damerell ladder, and the A17 numerological CKM hits (preserved from v7.4)

The arithmetic core of v7.4 – the integer-point Damerell ladder H_7' for $f_{4.5.\text{b.a}}$ at $K = \mathbb{Q}(i)$ – is unchanged in v7.5. We summarise the result for self-containment.

Theorem 1 (Bernoulli-anchored Cardy hit, v7.4 Theorem 2 reused)**.** *For the LMFDB CM newform $f_{4.5.b.a}$ (CM by $\mathbb{Q}(i)$, weight $k = 5$, level $N = 4$) and the Chowla–Selberg period $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi}) \approx 2.62205755$, the algebraic part $\alpha_2 := L(f_{4.5.b.a}, 2)\,\pi^2/\Omega_K^4$ equals $\alpha_2 = 1/12 = B_2/2 = -\zeta(-1)$, yielding the Cardy density $\rho_{\text{Cardy}}(c = 1) = c/12 = 1/12$ of the free boson parameter-free. The Γ-factor functional equation forces $\alpha_3 = \Gamma(2)/\Gamma(3) \cdot \alpha_2 = 1/24$, reproducing the $c = 1/2$ Ising central charge as a structural Γ-shadow. The full ladder $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} = \{1/10, 1/12, 1/24, 1/60\}$ is the standard Eisenstein–Kronecker ladder for a weight-5 CM-by-$\mathbb{Q}(i)$ newform [**?**, **?**, **?**, **?**, **?**, **?**].*

A17 numerological CKM hits (preserved with empirical caveat). The two A17 alignments of v7.4 remain on the v7.5 record:

- $|V_{us}| = (9/4)\,\alpha_1 = 9/40 = 0.22500$, matching PDG 2024 0.22501 ± 0.00068 to 0.015σ.
- $|V_{cb}|^2 = \alpha_1\,\alpha_4 = 1/600 \approx 1.667 \times 10^{-3}$, matching HFLAV $(1.668 \pm 0.057) \times 10^{-3}$ to 0.024σ.

The multiplicative coefficients $(9/4, 1)$ are flagged *empirical*, not structurally derived: without a flavour-symmetry quantum-number derivation of these coefficients, the rows are "thin coincidences" awaiting either group-theoretic interpretation or precision-experiment falsification (NA62 + FLAG-2027 by 2028 for V_{us}; LHCb-U2 / Belle II run 3 by 2030+ for V_{cb}). Cross-K on $K \in \{\mathbb{Q}(i), \mathbb{Q}(\sqrt{-2}), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-11})\}$ confirms $K = \mathbb{Q}(i)$ uniqueness (deviations 524%–15,490% on analogous ladders), but this K-specificity is essentially a consequence of the Bernoulli/Hurwitz alignment at $K = \mathbb{Q}(i)$, not independent SM evidence.

The unfit central charges $c = 7/10$ (tricritical Ising) and $c = 4/5$ (tetracritical Potts) of the unitary minimal series remain open: either they arise from the BDP p-adic anti-cyclotomic complement at $s = k/2$, or they require a second CM newform (the candidate is LMFDB 16.5.c.a, level 16, χ_4 nebentypus, paired with $f_{4.5.\mathrm{b.a}}$ in P-NT companion identifications of hatted weight-5 multiplets). Discriminating between these two options is the next concrete arithmetic test of v7.5.

7 Conditional analytical τ derivations: two H_6-co-dependent partial chains (A54)

Two analytical chains have been explored to derive the $\tau_S = i$ fixed point from arithmetic principles. Both are *partial* and *H_6-co-dependent* – they do not constitute "two independent analytical derivations". The v7.5 amendment records this honestly per A54 scoping.

Chain A: Galois descent on the $\Gamma(4)$ modular curve (P-NT). A bijective map between hatted weight-5 S_4' multiplets and a pair $(f_{4.5.\mathrm{b.a}},\ f_{16.5.c.a})$ of LMFDB CM newforms over $\mathbb{Q}(i)$ [**?**] forces $\tau_S = i$ via the unique $\Gamma_0(4)$-fixed point of the Galois action; the χ_4 nebentypus restriction H_6 selects $\alpha_2 = 1/12 = B_2/2$ as the canonical algebraic part. This is the chain written up in the companion P-NT paper (BLMS submission queue).

Chain B: Bost–Connes $\times$ CM at $\beta = 2\pi$ (A54 scoping). A new chain uses the Connes–Marcolli–Ramachandran KMS-states-$\times$-complex-multiplication framework [**?**] (`arXiv:math/0501424`) for imaginary quadratic K including $K = \mathbb{Q}(i)$. The five-step chain is: (S1) BC$\times$CM construction admits a $\beta = 2\pi$ formulation (✓ via CMR05); (S2) the 2-fold extremal-KMS ambiguity i vs. $-i$ for $h(\mathbb{Q}(i)) = 1$ (open); (S3, dominant obstruction) $\beta = 2\pi$ does NOT pick out infinity-type $(4, 0)$; the H_6 χ_4 nebentypus is still required as input – Chain B does *not* improve on Chain A's H_6-dependency; (S4) the type-$\mathrm{III}_1/\mathrm{II}_\infty$ obstruction reappears when identifying $\beta = 2\pi$ between the ECI-side DEHK [**?**] type-II_∞ algebra and the CMR05 type-III_1 algebra of BC$\times$CM (the bypass of A43 [**?**] is only PARTIAL); (S5) the Grössencharakter $\psi \to f_{4.5.\mathrm{b.a}}$ θ-lift via Hecke 1937 (✓).

Honest verdict. Chain B is genuinely new and publishable as a conditional math-ph note in 6–8 weeks sub-agent budget; full closure as an analytical theorem requires 24+ months and a co-author (Marcolli, Neshveyev). The corrected probability of closure as a clean derivation of $\tau = i$ *independent of* H_6 is $\sim 12\%$, down from the earlier Opus scoping estimate of 25%. v7.5 therefore records the chain as a SECOND H_6-co-dependent partial derivation, not a second independent one.

Brief-internal correction (logged not propagated). The earlier internal scoping had cited `arXiv:hep-th/9407124` as "Bost–Connes 1995" – this ID is in fact Gelfand–Krob–Lascoux–Leclerc–Retakh–Thibon "Noncommutative symmetric functions". The actual Bost–Connes 1995 paper (Selecta Math. NS 1.3, 411–457) is pre-arXiv. The chain's primary reference is CMR05 (`arXiv:math/0501424`), which generalises BC×CM to imaginary quadratic K including $\mathbb{Q}(i)$. No fabricated bibliography entry was produced; the brief-internal mis-tag is logged in A54 SUMMARY (hallu counter unaffected).

8 Falsifiers: seven experimental disambiguators 2027–2045

The v7.5 amendment is falsifiable on seven near-future experimental timescales, listed in chronological order with target year, central-value prediction, and confidence-level threshold for refutation.

1. **NA62 $K \to \pi\nu\bar{\nu}$ + FLAG-2027 lattice f_K/f_π on $|V_{us}|$ (2027–2028).** v7.5 prediction $|V_{us}| = 9/40 = 0.22500$ (Section **??**) tested at $\sim 0.3\%$ precision by 2028. $|V_{us}|$ outside 0.225 ± 0.001 at 3σ by 2028 $\Rightarrow$ **REFUTED** (this row). Empirical, not structural; survival margin ~ 0.5–1.5σ.

2. **KamLAND-Zen final $\langle m_{\beta\beta}\rangle$ + nEXO 2030+ (2027–2030).** v7.5 prediction $m_{\beta\beta} \sim$ few meV (CSD$(1+\sqrt{6})$ at NO with $m_1 = 0$, Section **??**). Current KamLAND-Zen upper bound $\langle m_{\beta\beta}\rangle < 28$–$122$ meV [**?**]. A measured $\langle m_{\beta\beta}\rangle > 50$ meV at 3σ from KamLAND-Zen 2027 / nEXO 2030+ refutes the NO + $m_1 = 0$ component of A14 $\Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row).

3. **JUNO 2030 NMO at 3–4σ (2030).** v7.5 prediction normal ordering with $m_1 = 0$ (Section **??**). Inverted ordering at $\geq 3\sigma$ by 2030 [**?**] $\Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row).

4. **DUNE 2030+ δ_{CP} (2030–2032).** v7.5 prediction $\delta_{CP} \approx -87°$ (Section **??**; CSD$(1+\sqrt{6})$). DUNE 2030+ at $\pm 15°$ [**?**] on δ_{CP} rules out the prediction at $> 3\sigma$ if δ_{CP} centres at 0 or $+90°$ $\Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row). Sensitivity sufficient by ~ 2032. The leptogenesis $(n-1)^2 = 6$ signature (Section **??**) inherits this falsifier indirectly through the joint $(\theta_{12,13,23}, \delta_{CP}, m_2, m_3, Y_B)$ global fit: if the seven-observable χ^2/dof prefers $n = 1+\sqrt{6}$ over $n = 3$ at $> 2\sigma$ by 2030+, A14 + A55 are corroborated; if χ^2/dof degrades by $> 3\sigma$ relative to $n = 3$, $\Rightarrow$ **REFUTED**.

5. **CMB-S4 Σm_ν at ~ 0.04 eV (~ 2032).** v7.5 prediction $\Sigma m_\nu \approx 0.06$ eV (CSD$(1+\sqrt{6})$ NO floor, Section **??**). CMB-S4 [**?**] 2032+ at $\sigma(\Sigma m_\nu) \sim 0.04$ eV discriminates NO + $m_1 = 0$ (preferred prediction 0.058 eV) from inverted with $m_3 = 0$ (0.10 eV) at $\sim 2\sigma$. A measured $\Sigma m_\nu > 0.10$ eV at 3σ $\Rightarrow$ **REFUTED** (CSD$(1+\sqrt{6})$ row).

6. **LHCb-U2 / Belle II run 3 on $|V_{cb}|^2$ (2030+).** v7.5 prediction $|V_{cb}|^2 = 1/600 \approx 1.667 \times 10^{-3}$ (Section **??**). Belle II 2027 at $\sim 0.5\%$ on $|V_{cb}|^2$ gives $\sim 0.12\sigma$ separation (insufficient). LHCb-U2 / Belle II run 3 (2030+) at $\sim 0.2\%$ precision discriminates: $|V_{cb}|^2$ outside $1.667 \times 10^{-3} \pm 5 \times 10^{-6}$ at 3σ by 2030 $\Rightarrow$ **REFUTED**.

7. **Hyper-K + DUNE proton-decay branching ratio (2030–2045).** v7.5 prediction (post-G1.12.B M_6) $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ at 95% CL (Section **??**). Outside $[0.3, 3]$ at $> 3\sigma$ by Hyper-K + DUNE combined 2030+ data, OR $\tau(p \to e^+\pi^0) > 3 \times 10^{35}$ yr at $> 2\sigma$ $\Rightarrow$ **REFUTED**. Decisive Hyper-K 20-yr measurement ~ 2045. This is the sole Nobel/Foundational-class testable prediction in the portfolio; G1.12.B M3–M6 closure (3.5–5 sub-agent months) is the immediate work-front.

To these we append a continuing v7.4 falsifier:

(8) NMC cosmological NULL at Cassini. A future improvement of the Cassini bound to $|\gamma - 1| \leq 10^{-6}$ that excludes the metric-NMC NULL falsifies the cosmological branch of v7.5-metric (but NOT v7.5-P; see Section **??**).

9 Outlook and open questions

We collect the open questions and conditional research directions that remain after v7.5.

Q1: dMVP26 quark CP-phase compensator (Section ??). Quantitatively, does a G1.12.B M3–M4 RGE matching reproducibly displace the modulus to $\mathrm{Re}\,\tau \sim 10^{-3}$ at the GUT scale, recovering the observed Jarlskog $J^q_{CP} \sim 3 \times 10^{-5}$ via the linear scaling of [**?**] p. 9? Alternatively, does the lepton-sector CP-spurion route (a) feed the CKM phase via a GUT-scale Yukawa-mixing term? This is a 12–16 week sub-agent campaign post-G1.12.B M3–M4 closure.

Q2: BDP vs. second-form discrimination of $c \in \{7/10, 4/5\}$ (Section ??). Either the BDP p-adic anti-cyclotomic complement supplies these unfit central charges via Iwasawa-algebra interpolation, or the v7.5 anchor must extend to a two-form anchor $(f_{4.5.\mathrm{b.a}}, f_{16.5.c.a})$. Discriminating (a) from (b) is the next concrete arithmetic test.

Q3: A54 BC×CM scoping memo (Section ??). 4-week Sonnet sub-agent to formalise the conditional math-ph note (publishable in 6–8 weeks if the chain is at least *conditionally* closed). Q1 2027 target.

Q4: v7.5-P Palatini sub-branch decision matrix (Section ??). The Palatini sub-branch makes four new predictions (P1–P4) under the assumption that the Wolf et al. 2025 large-ξ signal [**?**] is real:

- P1: under a Palatini posterior, $\xi_\chi \in [+1.5, +3.0]$, $\log B > +5$ vs. ΛCDM.
- P2: $G_{\mathrm{eff}}(z = 0)/G_N \in [1.4, 1.8]$ in DESI Y3 + LSST 2030.
- P3: $|\gamma - 1| < 10^{-9}$ at 1 AU (Palatini Yukawa screening).
- P4: all v7.5 arithmetic-modular falsifiers (DUNE δ_{CP}, Belle II V_{cb}, Hyper-K p-decay, $\mathrm{CSD}(1 + \sqrt{6})$, Damerell ladder) preserved unchanged.

Decision matrix:

- Wolf signal independently confirmed by 2027 (e.g. ACT DR6 + DESI Y3) ⇒ COMMIT v7.5-P as primary cosmological branch.
- Wolf signal refuted (e.g. ACT DR6 finds no 4σ enhancement) ⇒ DROP v7.5-P; v7.5-metric remains baseline.
- All v7.5 arithmetic predictions close NEGATIVE by 2030 ⇒ DROP entire programme; v7.5-P moot.

Q5: Leptogenesis sharper falsifier (Section ??). The $(n-1)^2 = 6$ signature is indirect; a sharper falsifier requires either a derivation of the high-energy phase η from ECI primitives (none known), or a cosmological probe of leptogenesis distinct from BBN Y_B. 4-6 week sub-agent campaign A56 in queue (NMC retrain Wolf-large-ξ + leptogenesis joint).

Q6: Modular-Higgs assignment as Axis-1 extension. The Higgs doublet H as an S_4' non-trivial representation, in the sense of Penedo–Petcov 2020 [**?**] or Kobayashi–Tanimoto, would constrain λ_H modular-form-wise. Cost $\sim$ 8–12 months sub-agent + Higgs-sector modular-form expert (Penedo, Petcov, King, Tanimoto). Foundational-Prize relevance LOW unless the prediction is at $< 5\%$ precision; current PDG M_W/M_Z precision is 0.01%, so any modular prediction is unlikely to be falsifiable as new physics. We register this as a deferred Q3 2026 outreach item.

Closing remark. The v7.5 amendment consolidates the v7.4 architecture, retracts the LYD20 unified-scaffold graft per A46, and integrates four well-scoped grafts (Karam-Palatini, KW dS-trap, dMVP26 Kähler, A55 leptogenesis) plus two structural reframings (A49 two-tau, A54 BC×CM honesty). The seven-falsifier table of Section **??** spans 2027–2045 across five major experiments (NA62, KamLAND-Zen, JUNO, DUNE, CMB-S4, Belle II/LHCb, Hyper-K). The arithmetic-modular core ($f_{4.5.b.a}$, $\mathbb{Q}(i)$, Damerell ladder, CSD$(1+\sqrt{6})$) is unchanged; the immediate work-front is G1.12.B M3–M6 closure (3.5–5 sub-agent months) toward the proton-decay branching ratio prediction at Hyper-K 2045+.

Acknowledgements

KR thanks the Sonnet sub-agent runs A46 (LYD20 graft retraction), A47 (KW dS-trap compatibility), A48 (dMVP26 Kahler at $\tau = i$), A49 (Du–Wang two-tau autopsy), A50 (Karam-Palatini Cassini reference), A54 (BC×CM at $\beta = 2\pi$ scoping), A55 (CSD$(1+\sqrt{6})$ leptogenesis), and A36 (G1.12.B M5 proton-decay partial widths), all conducted on 2026-05-05 under the parent Opus continuation thread. Computational resources: PC RTX 5060 Ti (cosmopower-jax 386 predictions/s, JAX named_shape patch). LMFDB queries: 4.5.b.a L-values cached 2026-05-05.

Data and code availability

Repository: `https://github.com/AIdevsmartdata/crossed-cosmos`. Snapshot of v7.4–7.5 axioms: Zenodo DOI `10.5281/zenodo.20030684` (commit 6.0.53.7, 2026-05-05). v7.5-specific evidence: A46 `A46_LYD20_GRAFT/lyd20_pinned_results.json` (99,061× penalty); A47 `A47_KW_DSTRAP_COMPAT/` (KW `arXiv:2310.10369` live-fetch); A48 `A48_DMVP26_KAHLER_TAU_I/` (`kahler_tau_i.py`, `diag.py`); A50 `A50_KARAM_PALATINI/` (`section2_patch.tex`, `v75P_sketch.md`); A54 `A54_BC_CM_BETA_2PI/SUMMARY.md` (CMR05 `arXiv:math/0501424` chain analysis); A55 `A55_LEPTOGENESIS/` (`lepto_eta_B.py`, `v75_section_lepto` `lepto_summary.json`); A36 `A36_G112B_M5/` (`proton_decay_widths.py`, `tau_p_results.json`). v7.4 baseline: cf. v7.4 amendment v2 [**?**].

References

[1] K. Remondière, *ECI v7.4 amendment (v2): τ as a CM-anchored attractor, the Damerell-ladder Cardy/CKM bridge H_7', and a CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at $K = \mathbb{Q}(i)$*, this volume (2026-05-05).

[2] K. Remondière, *A no-go theorem for the Cabibbo angle in S_4' modular flavour models at $\tau = i$*, ECI v7 aspiration note V2 (2026-05-04); companion paper, this volume.

[3] P. P. Novichkov, J. T. Penedo, S. T. Petcov, *Double Cover of Modular S_4 for Flavour Model Building*, Nucl. Phys. B **963** (2021) 115301; `arXiv:2006.03058`.

[4] X.-G. Liu, C.-Y. Yao, G.-J. Ding, *Modular invariant quark and lepton models in double covering of S_4 modular group*, Phys. Rev. D **103** (2021) 056013; `arXiv:2006.10722`.

[5] I. de Medeiros Varzielas, M. Paiva, *Quark masses and mixing from Modular S_4' with Canonical Kähler Effects*, `arXiv:2604.01422` (2026).

[6] X. K. Du, F. Wang, *Flavor Structures of Quarks and Leptons from Flipped SU(5) GUT with A_4 Modular Flavor Symmetry*, JHEP **01** (2023) 036; `arXiv:2209.08796`.

[7] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, I*, Acta Arith. **17** (1970/71) 287–301; *II*, Acta Arith. **19** (1971) 311–317.

[8] G. Shimura, *The special values of the zeta functions associated with cusp forms*, Comm. Pure Appl. Math. **29** (1976) 783–804.

[9] S. Chowla, A. Selberg, *On Epstein's zeta-function (I)*, Proc. Nat. Acad. Sci. U.S.A. **35** (1949) 371–374.

[10] A. Hurwitz, *Über die Entwicklungskoeffizienten der lemniskatischen Funktionen*, Math. Ann. **51** (1899) 196–226.

[11] G. Harder, N. Schappacher, *Special values of Hecke L-functions and abelian integrals*, in: *Workshop Bonn 1984*, Lecture Notes in Math. **1111**, Springer (1986) 17–49.

[12] R. Visser, *Special values of L-functions of CM modular forms*, University of Warwick first-year PhD project (2021).

[13] K. Remondière, *ECI/crossed-cosmos repository, version 6.0.53*, Zenodo (2026-05-05); DOI `10.5281/zenodo.20030684`.

[14] G.-J. Ding, S. F. King, X.-G. Liu, J.-N. Lu, *Modular S_4 and A_4 symmetries and their fixed points: new predictive examples of lepton mixing*, JHEP **12** (2019) 030; `arXiv:1910.03460`.

[15] I. de Medeiros Varzielas, S. F. King, M. Levy, *Littlest Modular Seesaw*, `arXiv:2211.00654` (2022).

[16] S. F. King, *Minimal predictive see-saw model with normal neutrino mass hierarchy*, JHEP **07** (2013) 137; 2016 follow-up `arXiv:1512.07531`.

[17] K. Remondière, *G1.12.B scoping: SU(5) $5_H + 45_H$ modular f^{ij} matrix campaign*, ECI v7 aspiration note A18 (2026-05-05); internal scoping document, this volume.

[18] N. Haba, K. Nagano, K. Shimizu, T. Yamada, *Gauge coupling unification and proton decay via 45 Higgs boson in SU(5) GUT*, PTEP (2024); `arXiv:2402.15124`.

[19] S. Antusch, J. Hinze, S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, `arXiv:2510.01312` (Oct. 2025; rev. Mar. 2026).

[20] A. Takenaka et al. (Super-Kamiokande Collaboration), *Search for proton decay via $p \to e^+\pi^0$ and $p \to \mu^+\pi^0$ with an enlarged fiducial volume in Super-Kamiokande I–IV*, Phys. Rev. D **102** (2020) 112011; `arXiv:2010.16098`.

[21] K. Abe et al. (Super-Kamiokande Collaboration), *Search for proton decay via $p \to \nu K^+$ using 260 kiloton-year data of Super-Kamiokande*, Phys. Rev. D **90** (2014) 072005; `arXiv:1408.1195`.

[22] K. Abe et al. (Hyper-Kamiokande Collaboration), *Hyper-Kamiokande Design Report*, KEK Preprint 2018-9 (2018); `arXiv:1805.04163`.

[23] B. Abi et al. (DUNE Collaboration), *Long-baseline neutrino oscillation physics potential of the DUNE experiment*, Eur. Phys. J. C **80** (2020) 978; `arXiv:2006.16043`.

[24] W. J. Wolf et al., *Cosmological constraints on a non-minimally coupled scalar field*, arXiv:2504.07679 (2025).

[25] S. de Vuyst, S. Eccles, P. A. Höhn, J. Kirklin, *Gravitational entropy is observer-dependent*, arXiv:2412.15502 (2024).

[26] K. Remondière, *Tensions audit and bypass-status table for the type-III_1/II_∞ identification across DEHK $\leftrightarrow$ CMR05 chains*, ECI v7 aspiration note A43 (2026-05-05); internal audit document, this volume.

[27] T. Chen, S. F. King, A. Medina, J. W. F. Valle, *Quark-lepton mass relations from modular flavor symmetry*, arXiv:2312.09255 (2023).

[28] F. Domingo, H. K. Dreiner, D. Köhler, S. Nangia, A. Shah, *A Novel Proton Decay Signature at DUNE, JUNO, and Hyper-K*, arXiv:2403.18502 (2024).

[29] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, Eur. Phys. J. C **82** (2024) 869; updated tables at http://flag.unibe.ch.

[30] A. Karam, S. Sánchez López, J. J. Terente Díaz, *Post-Newtonian Constraints on Scalar-Tensor Gravity*, arXiv:2604.16226 (2026), 42 pp. + 5 appendices.

[31] B. Bertotti, L. Iess, P. Tortora, *A test of general relativity using radio links with the Cassini spacecraft*, Nature **425** (2003) 374–376.

[32] S. F. King, X. Wang, *Modulus stabilisation in the multiple-modulus framework*, JCAP **07** (2025) 011; arXiv:2310.10369.

[33] S. F. King, A. Molina-Sedgwick, S. J. Rowley, *Fitting High-Energy Litebird Forecasts with the Littlest Seesaw*, JHEP **10** (2018) 184; arXiv:1808.01005.

[34] A. Connes, M. Marcolli, N. Ramachandran, *KMS states and complex multiplication*, Selecta Math. NS **11** (2005) 325–347; arXiv:math/0501424.

[35] S. Abe et al. (KamLAND-Zen Collaboration), *Search for Majorana Neutrinos with the Complete KamLAND-Zen Dataset*, arXiv:2406.11438 (2024).

[36] K. Abazajian et al. (CMB-S4 Collaboration), *Snowmass 2021 CMB-S4 white paper*, arXiv:2203.08024 (2022).

[37] A. Abusleme et al. (JUNO Collaboration), *JUNO Physics and Detector*, Prog. Part. Nucl. Phys. **123** (2022) 103927; arXiv:2204.13249.

[38] N. Aghanim et al. (Planck Collaboration), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641** (2020) A6; arXiv:1807.06209.